Same Inputs, Different Mechanisms: What Seven Time-Series Foundation Models Share

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Abstract

Time-series foundation models (TSFMs) are built on very different architectures, from decoder-only stacks over 32-step patches to encoder–decoders over quantized single time steps, yet little is known about whether they learn similar internal representations. We compare seven TSFMs on a common time axis measured from each model's token spans. The causal instruments are validated on a forecaster with planted concepts, and registered claims are tested once on sealed held-out splits of two independently generated corpora. All 21 model pairs share representational geometry above a shuffled-series null on held-out data, and features in unrelated architectures respond to the same input series. All 20 registered concept-transfer claims confirm on both corpora and on 800 raw real-world windows. The causal structure, by contrast, is largely model-specific. Causal features act mainly on the level and seasonality of the forecast, and features that raise level, lower level or damp seasonality occur in all seven models, but no fine-grained concept spans more than three of them, and the same effect on the same series is detected only between two Chronos models. Single features are also weakly specific, exceeding random directions drawn from the activation distribution only 1.29-fold. Several practices common in language-model interpretability give misleading results on these models. Averaging the tokens within a time window understates TimesFM's held-out CKA with fine-token models by 0.30–0.43, 83.5% of the most prominent dictionary features have no held-out causal effect, attention to seasonal lags does not predict which heads carry seasonality, and a plausible ablation null inflates the number of causal features 2.7-fold. Code, benchmarks and reproduction commands for every result are public.

Keywords: time-series foundation models, mechanistic interpretability, representational similarity, sparse autoencoders, causal intervention

1 Introduction

Pretrained forecasting models such as TimesFM (Das et al., 2024), the Chronos family (Ansari et al., 2024, 2025), Sundial (Liu et al., 2025), Timer (Liu et al., 2024) and Time-MoE (Shi et al., 2025) are released as open checkpoints and used zero-shot. They are built very differently. TimesFM reads 32-step patches with a decoder; Chronos-T5 quantizes every time step into a vocabulary and runs an encoder–decoder; Chronos-2 is an encoder with cross-series attention; Sundial samples its forecast from a flow-matching head; Timer reads 96-step tokens. Benchmarks rank them by accuracy but do not tell us what they represent. In vision and language, networks trained on similar data are argued to converge on similar representations (Li et al., 2016; Huh et al., 2024). It is not known whether forecasting models with such different inductive biases do the same, or whether any shared structure affects the forecast.

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We study this question for seven TSFMs. Three properties of forecasting models make the standard interpretability toolkit (Kornblith et al., 2019; Meng et al., 2022; Bricken et al., 2023) hard to apply directly, and in each case the failure is a plausible but wrong result rather than an error. Tokens cover between 1 and 96 time steps depending on the model, so activations must be placed on a common time axis before they can be compared. Real series carry no labels for trend, seasonality or noise, so a probe that appears to find seasonality cannot be checked on them. And removing a single internal feature typically moves a forecast by about a tenth of the context's standard deviation, small enough that a poorly chosen null or a precision mismatch can produce spurious effects. We therefore measure each model's token-to-time map, use generated benchmarks with exact ground truth, validate the causal instruments on a forecaster with planted concepts and decoys, and test registered claims once on sealed held-out splits of two corpora (Section 4).

Our results fall into three groups. The first concerns what the models share (Section 5). All 21 model pairs have representational geometry above chance, confirmed on a fresh sealed split, and concepts transfer across architectures at the level of inputs. For a concept found in one model, features in the other models select the same series, and all registered transfer claims confirm on both corpora and on 800 raw real-world windows that were never used during development.

The second concerns how the models differ (Section 6). Causal features act mostly on the level and seasonality of the forecast, and features that raise level, lower level or damp seasonality occur in every model. Finer-grained concepts, however, are organized by model, and none spans more than three of the seven. The same function can be implemented by one attention head in TimesFM and by five interacting heads in Chronos-T5-Base. Forecast-relevant information also appears to be distributed, since single features stand out only modestly against random directions drawn from the activation distribution. We detect the same causal effect on the same series only between Chronos-2 and Chronos-Bolt.

The third concerns methodology (Section 7). Averaging the tokens in a time window, the usual way of comparing models with different tokenizers, understates TimesFM's similarity to fine-token models by 0.30–0.43 CKA. Of the most frequently and strongly active dictionary features, 83.5% have no held-out causal effect. Attention to seasonal lags is uncorrelated with which heads carry seasonality, and a random-direction null that ignores a feature's per-token activation profile inflates the number of causal features 2.7-fold.

In addition, the depths at which TimesFM and Chronos-2 respond to input corruptions agree in a way that replicates for most corruptions, and internal features improve failure prediction only for the two least accurate models (Section 8). Each result is recorded, with its evidence class, exact numbers and a reproduction command, in a public findings ledger (`FINDINGS.md`); identifiers such as SH-18 refer to its entries. The analysis software and benchmark generator are open source (Appendix A).

2 Related Work

Interpreting TSFMs. Wiliński et al. (2025) study representational redundancy and concept steering in TSFMs, and four recent studies train sparse autoencoders (SAEs) on a single TSFM (Mishra, 2026; Oublal et al., 2026; Divo et al., 2026; Yıldırım, 2026); Kalnāre et al. (2025) apply patching and SAEs to time-series classifiers, and saliency methods for

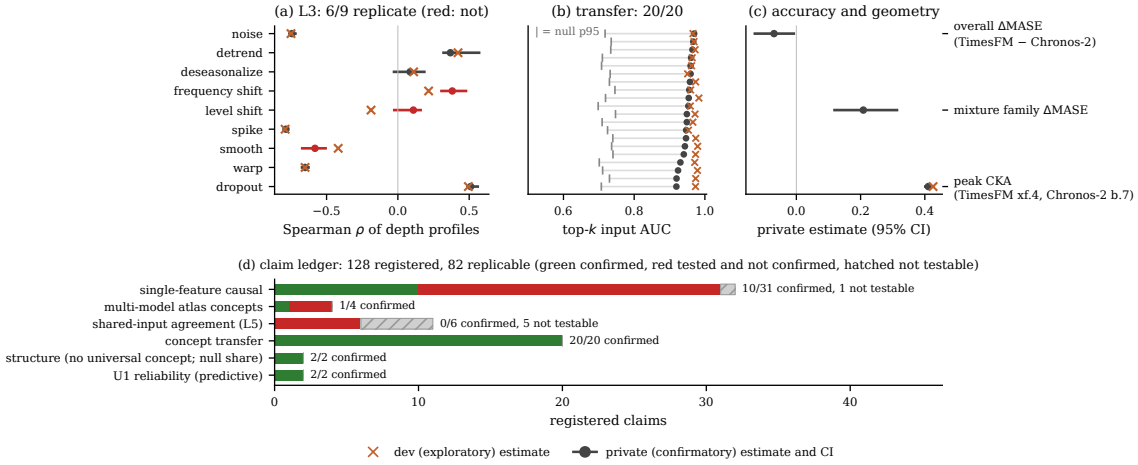


Figure 1: Claims re-tested once on the sealed v2 private split (971 series; panel a uses 800 of them). (a) Per-corruption depth-profile agreement, TimesFM vs. Chronos-2 (red: the dev value is not inside the private interval). (b) The 20 concept-transfer claims with private null 95th percentiles. (c) Accuracy and peak CKA; crosses are dev values, dots and bars private values with 95% series-bootstrap intervals. (d) Registered concept claims by type. Source: `confirm/confirmation.json`.

time-series classifiers are unreliable across architectures (Ismail et al., 2020). These works examine one model at a time and report features on the data they were found on. We compare seven architectures on a common time axis, separate agreement on inputs from agreement on effects, and test claims on held-out data. Our negative findings on activation prominence and on single-feature specificity bear directly on the single-model SAE studies.

Comparing representations. Representational similarity (Raghu et al., 2017; Morcos et al., 2018; Kornblith et al., 2019; Kriegeskorte et al., 2008) and model stitching (Lenc and Vedaldi, 2015; Bansal et al., 2021) underlie the convergence hypothesis for vision and language networks (Li et al., 2016; Huh et al., 2024). We use them only against shuffled-series nulls and report stitching as a gain over an input-feature baseline. Crosscoders (Lindsey et al., 2024) learn one dictionary across models; we instead compare independently trained dictionaries through the series they select and their effects on the forecast.

Validity of interpretability methods. Probes need control tasks (Hewitt and Liang, 2019; Belinkov, 2022); activation patching is sensitive to metric and corruption (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024); subspace interventions can create illusory mechanisms (Makelov et al., 2024); attention weights are not explanations (Jain and Wallace, 2019); and SAE features vary across seeds (Paulo and Belrose, 2025). Causal abstraction (Geiger et al., 2021) formalizes when an intervention supports a mechanistic claim. We address these concerns empirically with planted known answers, and borrow pre-registration (Nosek et al., 2018) and protection against adaptive holdout reuse (Dwork et al., 2015; Gelman and Loken, 2014) from empirical science.

Model	Checkpoint	Architecture	Token	Head	Params	Median MASE	FLOPs obs.
TimesFM 2.5	`google/timesfm-2.5-200m`	decoder-only	32	deterministic	231M	1.2560	42.5%
Chronos-2	`amazon/chronos-2`	encoder (+group attn.)	16	deterministic	119M	1.2002	97.5%
Sundial	`thuml/sundial-base-128m`	decoder + flow head	16	sampled	128M	1.4131	21.3%
Chronos-Bolt	`amazon/chronos-bolt-small`	encoder + patch head	16	deterministic	48M	1.5487	79.1%
Chronos-T5-Base	`amazon/chronos-t5-base`	encoder–decoder	1	sampled	201M	1.4231	13.9%[a]
Timer	`thuml/timer-base-84m`	decoder-only	96	deterministic	84M	1.9306	99.8%
Time-MoE	`Maple728/TimeMoE-50M`	decoder-only, MoE	1	deterministic	113M	2.8642	50.0%

Table 1: The seven models. "Token" is time steps per token; median MASE is on the v2 dev split (965 series; seasonal naive 1.9819); "FLOPs obs." is the share of a forecast's FLOPs spent in captured blocks at context 480. [a]Encoder only; the sampled decoder is not captured (DE-05).

3 Models and Data

Models. We analyze seven TSFMs (Table 1) spanning decoder-only, encoder-only and encoder–decoder stacks, token widths from 1 to 96 steps, a mixture of experts, and deterministic and sampled heads. Chronos-2 is the most accurate (median MASE 1.2002 on the v2 dev split, against 1.2560 for TimesFM and 1.9819 for the seasonal-naive forecast), and on the private split it is more accurate than TimesFM on the registered `mixture` family (ΔMASE 0.209 [0.119, 0.313], $p_{\text{Holm}} = 0.0002$, $n = 290$; v3 0.303). Time-MoE is less accurate than the naive forecast overall (median MASE 2.8642), so its internal results describe a model that does not forecast these series well. Per-model behavior is summarized in Appendix B.

Benchmarks with ground truth. Real series offer no ground truth for concepts, so every analysis runs on generated corpora in two labeled tiers. The synthetic tier touches no real data and records every generative component (trend, multiple seasonalities, AR noise, changepoints, anomalies, random walk, intermittency, heteroskedasticity). The real-derived tier (mixtures, block bootstrap (Künsch, 1989), sequential synthesis) inherits real source distributions and has attributional ground truth only. Every sample passes a leakage gate against the Monash archive (Godahewa et al., 2021) by banded DTW (Sakoe and Chiba, 1978). Each corpus has a dev split for exploration and a sealed private split from a disjoint seed range, hashed so that any change is detected.

- **v2** (primary): 965 dev series in five generator families, 410 of them real-derived (Monash weather, ETT (Zhou et al., 2021), electricity and traffic), and 971 private series.
- **v3** (replication): 2,876 dev series, 785 real-derived from 15 sources, and 1,024 private series. An external set of 800 raw windows from 12 GIFT-Eval datasets (Aksu et al., 2024), cut without any generator, re-tests the registered transfer claims, and a second sealed epoch tests the registered geometry claims.

Dev and private splits are exchangeable on both corpora (energy-distance $p = 0.951$ and 0.995; MP-09). Contexts have 480 steps and the horizon 64 steps.

Held-out confirmation. All results on a dev split are treated as exploratory. Selected dev results are frozen as hypotheses in a hash-pinned registry (v2: 128 hypotheses, 82 replicable on the private split; v3: 111 and 63) before the private split is generated, and each is tested once. Every interval and p-value resamples whole series, never windows (Efron

Registered claim type	v2 (971 series)	v3 (1,024 series)	Section
Accuracy (Chronos-2 better on `mixture`)	1/1	1/1	3
Peak-CKA pair	1/1	1/1	5.1
Concept transfer (shared inputs)	20/20	20/20	5.2
same claims on 800 raw real windows	–	20/20	5.2
Atlas structure	2/2	2/2	6.1
Single-model atlas concept	1/1	–	6.1
Cross-model causal (atlas concepts; agreement)	0/3; 0/6	none registrable	6.2
Single-feature causal effect	10/31	11/24	6.3
Perturbation-profile agreement	6/9	4/9	8
Failure prediction from internals	2/2	2/3	8
Confirmed / replicable (not testable)	43/82 (6)	41/63 (3)	
Window geometry, second sealed v3 split: 21 pairs above null; mean understates 4 pairs	–	21/21; 4/4	5.1

Table 2: Held-out confirmation on two independent corpora. Each cell is confirmed / tested; each claim type is one Holm family. The external raw-window set and the second v3 split (512 of its 2,863 series used) are not counted in the totals (CA-11, CA-12).

and Tibshirani, 1993; Cameron et al., 2008); each claim type is a Holm family (Holm, 1979). On v2, 43 of 82 replicable claims confirm; on v3, 41 of 63 (Table 2). The pattern is the same on both corpora. Claims about geometry and shared inputs replicate, single-feature causal claims replicate in part, and claims that two models share a causal effect do not. The following sections are organized along these lines.

4 How Models Are Compared

4.1 A common time axis

A 480-step context is 480 tokens in Chronos-T5 and Time-MoE, 30 in Chronos-2, Sundial and Chronos-Bolt, 15 in TimesFM and 5 in Timer, so token positions cannot be compared across models. We measure each token's time span I_j by injecting an impulse at every time step and accepting a span only if it passes contrast and contiguity gates (spans of all six hand-written adapters are recovered exactly from measurement; MP-01). Every model's tokens are then placed on one grid of windows $W_w = [wL, (w+1)L)$, $L = 32$ steps (Figure 2), and a window is represented by its tokens concatenated in time order,

$$\tilde{h}_w = \big[\, h_j \,\big]_{I_j \subseteq W_w}. \tag{1}$$

A token as long as a window maps to it one to one, and Timer's 96-step token is copied to the three windows it spans. Concatenation loses nothing, and linear CKA needs only the Gram matrix $K = \sum_p X_p X_p^\top$ summed over token positions, so the long vector is never formed. Alignment fixes the rows of two models' activations, not their coordinates, so every cross-model measure is defined on shared rows. Geometry compares how rows relate within each model, concept transfer compares which series a feature selects, and causal effects are compared on the forecast. Activations are never moved between models. During extraction an alignment gate checks, layer by layer, that an impulse in a window is answered most strongly by that window.

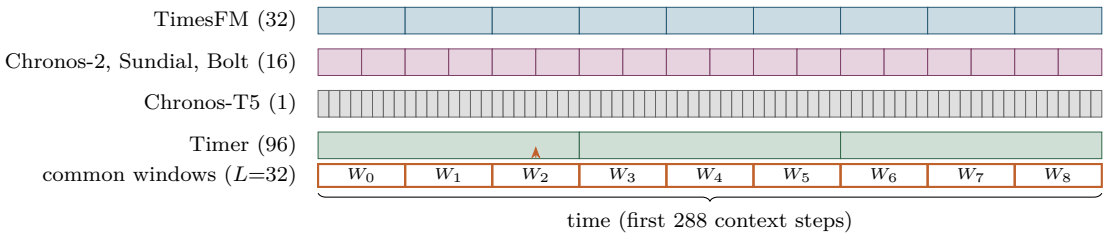


Figure 2: Time alignment. Tokens cover measured spans of different widths (Chronos-T5's one-step tokens are drawn in groups of four). Equation (1) places them on common 32-step windows: shorter tokens are concatenated in time order, TimesFM's tokens map one to one, and each Timer token is copied to the three windows it spans.

4.2 Geometry

For aligned, centered window representations A and B over the same series, linear CKA (Kornblith et al., 2019), $\mathrm{CKA}(A,B) = \|B^\top A\|_F^2/(\|A^\top A\|_F\|B^\top B\|_F)$, is computed for every layer pair of every model pair and read against a null that permutes whole series (1,000 permutations). Ridge stitching (Bansal et al., 2021) is reported only as its gain over a map from input features, since much of any layer is linearly predictable from the input alone.

4.3 Causal features

Dictionaries and ablation. Each target layer ℓ receives a TopK sparse autoencoder (Makhzani and Frey, 2013; Gao et al., 2025), $z(h) = \mathrm{TopK}(W_e(h-b_d)+b_e)$, $\hat{h} = W_d z(h)+b_d$, with three replicate seeds. Feature i is tested on the k series on which it fires most strongly by comparing the forecast under the reconstruction with $z_i = 0$, $\hat{h}_{-i}$, against the forecast under the full reconstruction,

$$\Delta_i^{(c)}(x) = g_c(f(x; h^{(\ell)} \leftarrow \hat{h}_{-i})) - g_c(f(x; h^{(\ell)} \leftarrow \hat{h})), \tag{2}$$

on nine forecast channels g_c: trend, seasonal strength, spectral centroid, level, dispersion, near- and far-horizon shape distance, MASE and flatness. Using the reconstruction in both arms keeps the dictionary's own error out of the comparison. Before any feature is scored, a reach probe requires that patching a layer into itself changes the forecast by exactly 0.0 and that patching another layer changes it, and sampled models are seeded identically in compared calls.

Random-direction nulls. A feature counts as causal on channel c only if its mean $|\Delta_i^{(c)}|$ exceeds the 95th percentile of removing the same amount of activation, token by token, along 16 random directions on the same series. Two direction distributions are used. The isotropic null (v2) draws directions uniformly in the dictionary's code space and decodes them. The covariance-matched null (v3) draws them from a Gaussian with the layer's own activation covariance, so they lie in the subspace the model actually uses. A feature's effect vector $v_i^{(c)}$ is its signed mean effect in units of its null's 95th percentile. Because a mean is compared with a percentile of single rows, the chance rate of clearing is estimated per cell by scoring each null draw against the others.

From features to concepts. Within a layer, causal features are clustered by effect vector (k-means, silhouette, at least three members per cluster). Across models, the atlas pools all causal features and cuts a complete-linkage dendrogram at cosine 0.9, so an atlas concept is a tight causal-effect profile; coarser families (average linkage at cosine 0.5) serve as a reading vocabulary.

4.4 Two kinds of sharing

We distinguish two senses in which a concept can be shared. For shared inputs, concept transfer asks whether some feature in another model separates a concept's top-k series from the rest (AUC), against stratum-matched random series sets; it is run in both directions with Benjamini–Hochberg control (Benjamini and Hochberg, 1995). For the same effect on the same inputs, both sides of every reciprocal transfer are ablated on the same series. Each side must clear its own null, and their level concordance and shape cosine must exceed floors obtained by ablating random feature sets matched to each side's activation. A statistic below both floors' 5th percentile is taken to mean that the two act differently. Failing to exceed the 95th percentile is not by itself evidence of a difference.

4.5 Validation on known answers

Because a misspecified null does not produce an error, each instrument was first run on a forecaster with planted concepts. One block of this forecaster writes known directions into the residual stream on series with a given property, and the head applies a known effect on level, trend, seasonality or dispersion. Decoys include input-only directions with no effect, sub-null directions with a weak effect and a control layer with none; for cross-model tests, two architectures with rotated residual bases share planted concepts. Pass criteria were fixed in advance and evaluated on five construction seeds (Table 3). Both nulls used in the study pass. The test does reject methods. A plausible null that removes one batch-averaged amount instead of each token's own profile fails on every seed and, on a four-model development run, inflates the number of causal features from 110 to 293 (MN-29). The same-effect test never calls a decoy "same" but rarely detects a concept two architectures truly share (MN-30). Its positive verdicts can therefore be trusted, but the absence of a positive verdict is not informative.

5 Shared Structure Across Architectures

5.1 Representational geometry

With concatenated windows, peak CKA (the maximum over layer pairs) between TimesFM and each other model is 0.358–0.635 on v2 and 0.375–0.712 on v3 (512 series each), and all 21 model pairs exceed their shuffled-series null on both corpora ($p = 0.001$, 1,000 permutations; null 95th percentiles 0.003–0.51, highest for pairs with Time-MoE). The layer pairs selected on v3 dev were registered and tested once on a fresh sealed split, and all 21 exceed their nulls (Holm $p = 0.021$), with TimesFM/Chronos-2 at 0.707 [0.679, 0.734] against 0.712 on dev (SH-01). Models with different tokenizers, depths and training objectives therefore organize series in measurably similar ways, which parallels the convergence reported for vision and language networks (Huh et al., 2024).

Instrument	Result on the planted forecaster	Pass criterion	Verdict
Isotropic null (v2)	false-positive rate $\leq$0.003, sensitivity 0.70–1.00, all 5 seeds	FPR $\leq$0.10, sens. $\geq$0.5	pass
Covariance-matched null (v3)	sensitivity 0.916, false positives in 5 of 2,948 control cells	FPR $\leq$0.10, sens. $\geq$0.5	pass
Batch-averaged null (reference)	false-positive rate 0.168–0.248 on every seed	FPR $\leq$0.10, sens. $\geq$0.5	fail
Same effect on same inputs	false "same" 0 of 9 decoys; detects a shared one-direction concept in 0.4 of seeds	–	specific, insensitive

Table 3: Known-answer tests of the causal instruments, with criteria fixed before each test (MN-29, MN-30).

Much of this similarity is lost under the usual comparison. Averaging the tokens within a window, which is the common way to compare models with different token widths, lowers TimesFM's held-out CKA with Chronos-2, Chronos-T5-Base, Chronos-Bolt and Sundial by 0.322, 0.433, 0.301 and 0.303 (each 95% interval above 0.26; Holm $p = 0.0008$; MN-38), confirmed for all four pairs on the second sealed v3 split. Fine-token models encode the order of time steps within a window, which a coarse-token model represents inside a single token, and averaging discards it. In the first block of Chronos-T5-Base, a window's slope decodes with $R^2 = 0.13$ from the window mean and 0.96 from finer sub-windows. Pairs with Timer change little, because its 96-step token has no within-window order to match. Linear stitching shows the same effect. Concatenation raises the gain over the input baseline, from 0.498 to 0.554 on average and from 0.307 to 0.788 for Chronos-T5-Base$\rightarrow$TimesFM, while keeping the ranking of pairs (Spearman 0.92).

5.2 Shared inputs

Geometry compares whole layers. Concept transfer is more specific. A concept is defined by the series on which one model's causal feature fires, and we ask whether another model contains a feature that singles out the same series. On v2 dev, 1,296 of 1,584 atlas transfer tests are reciprocal after Benjamini–Hochberg correction. Twenty transfer claims were registered from two source concepts, TimesFM layer 18 (14 claims) and Sundial layer 6 (6 claims), with destinations in all seven models. All 20 confirm on the 971 private series at the Holm floor attainable for 20 claims at 2,000 null draws (0.0100), with AUCs of 0.919–0.970 against null 95th percentiles of 0.70–0.75 (Figure 1b). On v3, a new set of 20 claims from three source models (Sundial 14, Chronos-2 5, Time-MoE 1) confirms 20 of 20 on the private split and 20 of 20 on the 800 raw GIFT-Eval windows, with AUC 0.757–0.991 (SH-18). The result does not depend on the alignment grid. Re-pooled at 64 and 96 steps, the v3 reciprocal rate moves only from 0.827 to 0.801 and 0.799 (SH-23). Only 8 of the 20 best-matching private features on v2 are the ones selected on dev, so what is confirmed is that the destination layer contains a feature selecting the same series, rather than the identity of that feature.

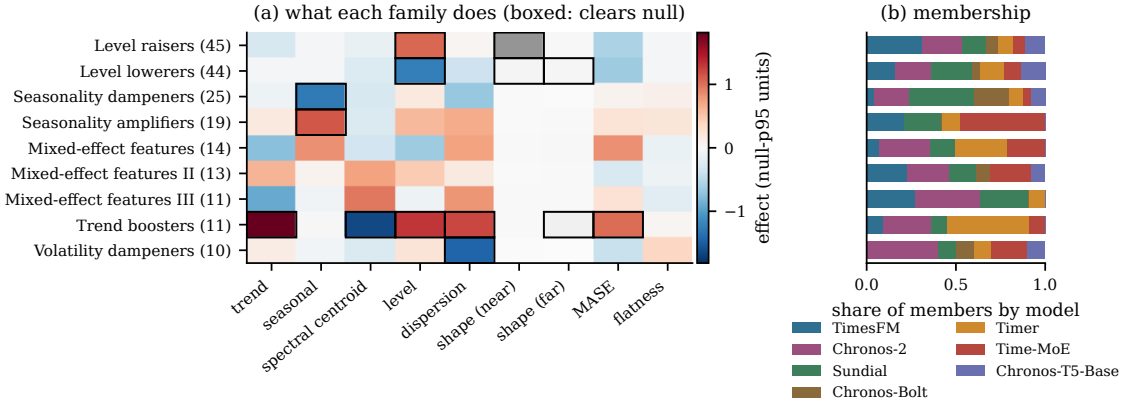


Figure 3: The nine causal-effect families. (a) Family centroid in null units, signed as what the feature does (red raises, blue lowers; boxed cells clear the null; shape distances unsigned, gray). (b) Membership by model. Source: `sae/concept_families.json`.

5.3 Common effect types

The causal features of all seven models fall into a small number of effect types (Figure 3). Nine families cover 192 of the 194 v2 causal features. The two largest act on level, namely level raisers (45 features) and level lowerers (44), together 89 of the 192. Seasonality dampeners (25) and amplifiers (19) follow; three mixed-effect families (14, 13 and 11), trend boosters (11) and volatility dampeners (10) are smaller. Level raisers, level lowerers and seasonality dampeners each have causal members in all seven models, and volatility dampeners in six. Since membership is determined by each feature's own measured effect, this does not depend on where the family boundaries are drawn. Together with the transfer results, it suggests that the seven models respond to the same properties of a series, mainly its level and seasonality, and adjust the forecast in the same directions.

6 Model-Specific Causal Structure

Shared inputs and similar broad effects do not imply that the models share mechanisms, that is, the same concept with the same effect on the same series. At this level of detail we find little sharing.

6.1 Concepts are organized by model

The atlas groups causal features from all models whose effect profiles agree at cosine 0.9. It finds more structure than a column-shuffle null (11 concepts, null 95th percentile 5.05; $p = 0.005$), but of the 11 concepts, none spans more than 3 of the 7 models. The clusters are purer by model than random assignment makes them: 9 multi-model concepts against a null mean of 10.8 ($p_{\text{below}} = 0.025$), mean model purity 0.591 against 0.474 (SH-22). Five of the 11 are convergent, meaning that two models produce the same effect on different series. The pattern holds on held-out data, where no concept of the private atlas spans more than

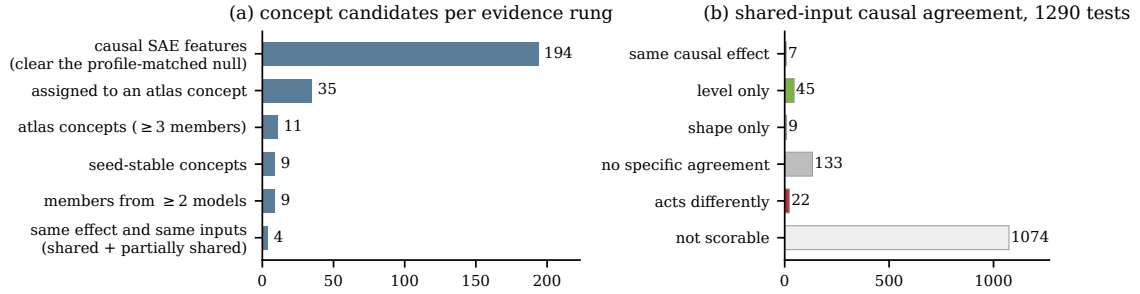


Figure 4: (a) Concept candidates at each rung (the first three bars are sequential; the last three are properties of the 11 atlas concepts). (b) Same effect on the same inputs, over 1,290 tests drawn from the 1,296 reciprocal transfers. Source: `sae/concept_stage.json`.

3 of 7 models on v2 or more than 2 on v3 (SH-16). Each architecture appears to build its own variants of level and seasonality effects.

A component-level example comes from seasonality. In an earlier two-model run, restoring a deseasonalized context required a single attention head in TimesFM but a minimal set of five heads in Chronos-T5-Base, whose contributions do not add (path-patching gap 0.0238 against a mean effect of 0.0108; CA-03; exploratory).

6.2 Agreement of causal effects

The strictest test of sharing ablates both sides of a reciprocal transfer on the same series. Of 1,290 tests, 216 are scorable (Figure 4b): 7 show the same causal effect, 54 agree on level or shape alone, 133 show no specific agreement and 22 act differently. All 7 "same" verdicts are between Chronos-2 and Chronos-Bolt, two models from one family, six of them through one feature set in Chronos-Bolt's fourth block. The test produced no false "same" verdicts on the planted forecaster, so we regard these detections as reliable. It is also insensitive, so the lack of detections across families does not by itself show that effects differ, although the 22 "acts differently" verdicts indicate that they sometimes do. Consistent with this, no registered cross-model causal claim confirms held-out. Three multi-model atlas concepts exceed the null in centroid cosine on private data (0.955–0.988 against 95th percentiles 0.854–0.907) but fail the registered member-pair rule, and 0 of 6 shared-input agreement claims confirm.

6.3 Specificity of single features

Causal features. Of 56 target layers, 55 pass both reach controls. Across them, 442 feature-channel cells clear the isotropic null where the empirical chance rate predicts 178.75 (2.47 times chance), from 1.64 times (Chronos-T5-Base) to 5.70 (Chronos-Bolt). Timer is the exception. Its layers reach the forecast, but its features clear at 1.02 times chance (PM-17). Target by target, 22 of 55 targets clear significantly more cells than chance, at least two in every model except Timer. Registered single-feature effects replicate only in part. Of 31 testable claims, 10 confirm on the v2 private split, each at the Holm floor for 31 tests ($p_{\text{Holm}} = 0.031$), and 11 of 24 on v3.

Comparison with on-manifold directions. The v3 run scored every feature against both nulls. Under the isotropic null, 238 of 967 scorable features clear at least one channel; under the covariance-matched null, 91 do. Scoring each null draw as a pseudo-feature on all nine channels at once gives expected counts of 67.0 and 70.4, so features exceed chance 3.55-fold against isotropic directions ($p = 4 \times 10^{-86}$) but only 1.29-fold against directions drawn from the activation distribution ($p = 0.0045$). Removing a typical on-manifold direction thus moves the forecast nearly as much as removing a dictionary feature. The effects are also small. The median single-feature effect is 0.097 context standard deviations, 2.2 times smaller than the 0.214 by which substituting the reconstruction moves the forecast (v3: 1.7 times; MN-15).

Other interventions. We re-tested the 91 v3 features that clear the on-manifold null under three on-manifold interventions, each with its own matched null. These replace the feature by its corpus mean, by its activation on another series where it fires, or subtract it from the model's own activation rather than the reconstruction. Of the 91, 79 clear under at least one and 37 under all three, and mean and resample ablation preserve the direction of the effect (median cosine to zero ablation 0.997 and 0.991; MP-10). Most of these features therefore have a genuine effect, but which of them clear depends on the intervention.

Absence of discrete units. Within a layer, 50 of 56 targets are non-modular, meaning that no choice of k splits their effect vectors into clusters of at least three members each. The families of Section 5.3 cover 192 of 194 features but do not beat a column-shuffle null in number ($p = 0.68$), and the silhouette is 0.25–0.32 for every cut from 6 to 16 families. The effect vectors are not low-dimensional (participation ratio 5.58 of 9 channels on v3; MN-07), so the lack of clusters is not due to a lack of variation. Together with the on-manifold comparison, this indicates that forecast-relevant information is spread over many directions and varies continuously. This differs from the monosemantic units sought in language models (Bricken et al., 2023), and it means that claims about single TSFM features should be tested against an on-manifold null.

7 Pitfalls in Transferring Interpretability Methods

Several practices that are standard in language-model interpretability give misleading results on TSFMs. In each case below, the problem was identified by a control and would otherwise have gone unnoticed.

Window pooling. As shown in Section 5.1, averaging the tokens within a window understates TimesFM's held-out similarity to four fine-token models by 0.30–0.43 CKA. It also changes which models appear similar. With mean-pooled representations, TimesFM/Chronos-2 ranks 15th of 21 pairs by CKA, although it is the most similar pair by error agreement (0.948); with concatenated windows it ranks 9th on v2 and 7th on v3. Comparisons of TSFM representations therefore depend on a pooling choice that is rarely reported.

Activation prominence. SAE analyses usually start from the features that fire most often or most strongly. In TSFMs these are mostly inert. Of the 927 most prominent features of the v2 private atlas, 0.835 have no causal effect on the forecast (one-sided lower bound 0.817; dev 0.810; $p_{\text{Holm}} = 0.0005$; MN-14), and on v3 the share is 0.919 of 861.

Ranking by measured effect does much better. Zeroing a layer's five attribution-ranked atoms produces a median 0.74 share of the forecast change the dictionary can express, more than activation-matched random sets of five atoms on 30 of 31 targets (D-42C).

Attention patterns. The attention a head places on seasonal lags is uncorrelated with the seasonal power lost when the head is ablated ($\rho = 0.026$ in TimesFM and -0.118 in Chronos-T5-Base; CA-04). This extends the observation of Jain and Wallace (2019) to forecasting models.

Correlation with generative factors. Because the benchmark has exact ground truth, features can be matched to generative factors. After dead-feature revival, 154 of 10,240 TimesFM features and 115 of 6,144 Chronos-T5-Base features correlate with a generative field beyond a permutation null, most strongly with intermittency ($\rho = 0.881$; MP-06). The correspondence is not specific, however. Varying the generator parameter along a dose ladder moves the matched feature no more than random features of matched activation, and none of 151 registered tests survives correction (MN-17). In an earlier exploratory run, steering the feature best matched to seasonal amplitude changed the forecast's seasonality in the expected direction in TimesFM and in the opposite direction in Chronos-T5-Base, in two independently trained dictionaries per model (CA-05).

Choice of null. Whether a feature counts as causal depends on the random intervention it is compared with, and reasonable choices differ severalfold. A null that removes one batch-averaged amount instead of each token's own amount inflates the number of causal features 2.7-fold (110 to 293) and fails the planted-forecaster test on every seed (Section 4.5). Between the two nulls that pass, isotropic directions and directions drawn from the activation distribution, the excess over chance falls from 3.55-fold to 1.29-fold (Section 6.3).

Similarity metrics. Seven pairwise similarity metrics, covering geometry, error agreement and concept transfer, rank the 21 model pairs inconsistently (Kendall's $W = 0.25$, $p = 0.0095$; SH-05, SH-06). Geometry also does not reflect lineage. Two untrained Chronos-T5 models are more similar (CKA 0.878) than the trained and related Chronos-T5-Small and Base (0.734; MN-01). We therefore report each metric against its own null and do not combine them into a single score.

8 Depth and Practical Value

Depth profiles. For each of nine input corruptions, a model's sensitivity fingerprint is the relative change of its residual stream at each layer; two models are compared by the rank correlation of their fingerprints on a shared relative-depth axis; activations are never moved between models. For TimesFM and Chronos-2 the agreement replicates on the v2 private split for 6 of 9 corruptions (overall $\rho = 0.524$ [0.504, 0.544]; Figure 1a) and for 4 of 9 on v3, with 8 of 9 keeping their sign. The two models react at matching depths to detrending (private $\rho = 0.368$) and dropout (0.510), and at opposite depths to noise (-0.743), spikes (-0.785) and warps (-0.651). Frequency shift (0.381 against 0.216 on dev) and smoothing (-0.581 against -0.418) fail the registered rule because the agreement is stronger on private data than on dev. Only for level shift is the agreement unstable; it changes sign between dev and private on both corpora (CA-01).

Window-level restoration. Patching clean activations window by window into a corrupted TimesFM pass shows that the forecast is restored mainly in the window just before the horizon, increasingly so with depth (from 0.427 of the damage undone at layer 0 to 0.612 at layer 8 and 0.631 at layer 16 for noise), while restoration in the first window falls from −0.191 to 0.000 (320 series; DE-01). This recency effect is not visible when whole layers are patched.

Comparability of depth. The captured blocks account for 13.9% (Chronos-T5-Base, encoder only) to 99.8% (Timer) of a forecast's FLOPs (DE-05), and on a whole-stack depth axis Chronos-T5's last encoder block sits at depth 0.478 (DE-04). Which layers appear most important also depends strongly on the selection criterion (DE-08). We therefore compare depth profiles across models rather than the locations of single layers.

Failure prediction. For each model we predict a series' log MASE from free, output-only features (quantile width, forecast flatness, catch22 features of the context; Lubba et al., 2019), add dictionary-family activations, and measure the gain, cross-fitted with a series bootstrap. The free baseline is strong (Spearman 0.76–0.85), and for five models the gain's interval includes zero. The gain is positive for the two least accurate models, Timer (+0.030 [0.019, 0.043]) and Time-MoE (+0.012 [0.005, 0.019]), and both confirm on the private split (Timer 0.0446 [0.0292, 0.0605], $p_{\text{Holm}} = 0.002$; Time-MoE 0.011 [0.0032, 0.0194], $p_{\text{Holm}} = 0.012$) and again on v3 (MN-28). Internals do not help in choosing a model. Routing each series to the model with the lowest predicted MASE gives 1.964 [1.837, 2.093] with internals against 1.948 [1.814, 2.070] without, both worse than always using Chronos-2 (1.856). For the more accurate models, the model's output carries as much information about its errors as its internal features do.

9 Discussion

The seven models differ in tokenization, depth and training objective, yet they select the same series for a concept, share whole-layer geometry, and contain features with similar broad effects on the level and seasonality of the forecast. The similarity largely ends at the level of individual causal effects. Fine-grained concepts are organized by model, the same function can be carried by one head or by a five-head circuit, and the same effect on the same series is detected only within one model family. One interpretation is that the models converge on which properties of a series are predictive, which is largely determined by the data, while the way the forecast responds to them depends on the architecture. This is a weaker form of convergence than that proposed for vision and language models (Li et al., 2016; Huh et al., 2024). With the present instruments, agreement on inputs is measurable and replicates, whereas agreement on effects across model families lies below the resolution of single-feature ablation at these effect sizes.

The results also have practical consequences for interpreting TSFMs. Single dictionary features are only weakly specific relative to directions the model actually uses, so single-feature claims, including those of recent single-model SAE studies (Mishra, 2026; Oublal et al., 2026; Divo et al., 2026), should be evaluated against an on-manifold null. Feature importance has to be established by intervention, since activation prominence, attention and correlation with a generative factor each identify the wrong components. Cross-model

comparisons need a measured time axis and a pooling that preserves the order of time steps within a window. Finally, with thousands of exploratory comparisons (Gelman and Loken, 2014), registration and a single held-out test were essential. They confirmed between half and two thirds of the registered claims (43 of 82 and 41 of 63), and the claim types separated consistently into those that replicate and those that do not.

10 Limitations

Models and alignment. The study covers seven transformer models with contiguous tokens; models whose tokens read disjoint lags, such as Lag-Llama, are out of scope. With seven models, differences between models are confounded with architecture, size and training data, and Timer's features do not exceed chance, so its results carry no evidence about causal concepts. The analysis is univariate (Chronos-2's cross-series attention is not measured; PM-13), and Chronos-T5's decoder and Sundial's flow head are not captured. The common time grid is no finer than the coarsest token, and the locations of peak layer pairs, though not the size of the similarity, depend on the window size (SH-23). Dictionaries are trained on window vectors, which is conservative for one-step models (MP-11).

Ground truth and scope. Ground truth is exact only for the synthetic tier and encodes the generators' assumptions, and overlap with the models' pretraining data cannot be verified. Raw real benchmarks are a weaker basis for causal questions, because weakly predictable real series receive flat forecasts in 67–100% of cases (BM-06; cf. Wan et al., 2026), leaving an ablation nothing to move. Real data therefore enter as real-derived series and as the external test set on which the transfer claims confirm. The confirmed transfers derive from two source concepts on v2 and three source models on v3, so the shared-input result rests on few concepts. Concept results depend on dictionary size, sparsity and seed; three replicate seeds measure this dependence but cannot remove it.

11 Conclusion

We compared seven time-series foundation models on two independent corpora, using causal instruments validated on planted concepts and testing registered claims once on sealed data. Architectures that differ substantially share representational geometry, select the same inputs for a concept and contain features that raise or lower the level of the forecast or damp its seasonality. Their causal structure, however, is model-specific and distributed. Several practices adopted from language-model interpretability, including averaging tokens, ranking features by activation and reading attention patterns, give misleading results on these models. The benchmark, the analysis software and the reproduction commands for all results are public, so the conclusions can be re-tested on new models.

Availability. Code (MIT license), configurations, the example run and the findings ledger are at `https://github.com/DanK10097110/TSFM-Interp`. Version 1.1.0 is archived at doi:10.5281/zenodo.23267462, and both packages install from PyPI (`pip install tsfm-lens tsfm-benchmark`).

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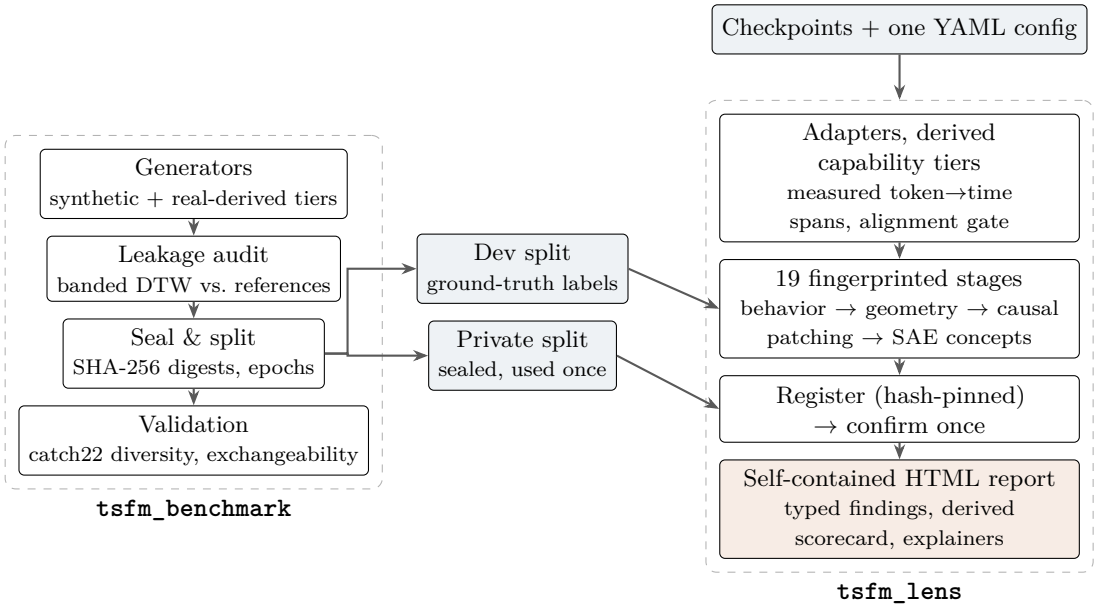


Figure 5: System overview. `tsfm_benchmark` produces leakage-audited, sealed corpora with exact ground truth; `tsfm_lens` turns a configuration naming checkpoints into one report. Dev findings are exploratory; the private split is opened once, by `confirm`, for claims registered in advance.

Computational resources were provided by the second author's lab at Rutgers University. This work received no external funding, and the authors declare no competing interests. Use of AI tools: Claude (Anthropic) wrote most of the repository's code and tests, ran the experiments and helped draft and edit the manuscript, under the authors' direction. The research questions, ideas and general design choices are the authors', who reviewed the implementation details Claude proposed, checked every reported result and take full responsibility for the content.

Appendix A. The Analysis System

All analyses are produced by TSFM-Lens, which consists of two separately installable packages. `tsfm_benchmark` generates, leakage-audits, seals and validates corpora; `tsfm_lens` loads checkpoints through adapters, runs a gated pipeline of 19 analysis stages and renders one self-contained HTML report (Figure 5). A zero-code adapter for Hugging Face checkpoints discovers layers and measures token spans, and loads two of the seven models here (Timer and Time-MoE). Stages a model cannot support are skipped with a reason rendered in the report; every intervention goes through one patching primitive, which raises if the model silently splits a batch; and the `confirm` stage refuses mismatched registry hashes, requires the private seal to verify and refuses to overwrite its own artifact. The system and its design are described in a companion software paper; usage is documented in the repository.

Model	median MASE	80% coverage	scale resid. (s.d.)	missing value	latency (ms)
Chronos-2	**1.2002**	0.757	0.0102	handled	23
TimesFM	1.2560	**0.779**	1.6×10^{-6}	propagates	39
Sundial	1.4131	0.360	0.0310	propagates	62
Chronos-T5-Base	1.4231	0.436	2.7×10^{-7}	handled	975
Chronos-Bolt	1.5487	0.732	0.0101	handled	15
Timer	1.9306	–	0.0262	propagates	5
Time-MoE	2.8642	–	0.0368	propagates	103
naive / seasonal naive	2.7593 / 1.9819				

Table 4: Behavior on the v2 dev split (965 series). Scale residual: error of $f(cx)/c$ against $f(x)$ in context standard deviations, $c = 0.001$. Missing value: whether an injected NaN is handled or propagates silently. Latency: median time to forecast a batch of 8 series. Timer and Time-MoE output point forecasts only.

Appendix B. Model Behavior

Table 4 summarizes behavior on the v2 dev split. Five models beat the seasonal-naive forecast on 76.5–84.5% of series, Timer on 69.5% and Time-MoE on only 37.8%. The nominal 80% intervals of Sundial and Chronos-T5-Base cover only 36% and 44% of outcomes, and four models propagate an injected missing value silently. On the private split, Chronos-2 is more accurate than TimesFM overall, with a smaller effect than on `mixture` (-0.070 $[-0.129, -0.008]$, $p = 0.025$, $n = 971$). Sampled models (Sundial, Chronos-T5) are seeded identically in every compared pair of calls: unseeded, back-to-back Sundial calls differ by up to 0.50 MASE (PM-09).

Appendix C. Statistical Details and Validity

Paired series bootstrap. With per-series differences $d_s = m_{a,s} - m_{b,s}$ and $B = n_{\text{boot}} = 5000$ resampled means $\bar{d}^*_b$, the interval is the percentile interval and $p = \min(1, \max(\frac{1}{B}, 2\min\{\frac{1}{B}\#(\bar{d}^*_b \le 0), \frac{1}{B}\#(\bar{d}^*_b \ge 0)\}))$ (Phipson and Smyth, 2010). Window-level statistics (CKA, fingerprints) resample series with all their windows (Cameron et al., 2008).

Multiplicity. Confirmatory families use Holm, $\tilde{p}_{(i)} = \max_{j \le i} \min(1, (m-j+1)p_{(j)})$ (Holm, 1979); with $p \ge 1/B$ the smallest attainable value is m/B, which is checked per family. Atlas transfer uses Benjamini–Hochberg (Benjamini and Hochberg, 1995). The v2 run has 123 test families with 4,238 comparisons; the 8,705 feature-channel cells of the ablation battery are judged by their count against the empirical chance rate, which ranged from 0.3% to 4% per target.

Ablation nulls. For feature i with rows R_i and $J = 16$ unit directions u_j, the null removes at every token t the amount $|z_i(t)|\,\|w_i\|\,u_j$. In the isotropic null u_j is $W_d^\top r_j$ normalized, with r_j a random unit code; in the covariance-matched null $u_j \propto \sum_t g_{jt}(x_t - \bar{x})$ with $g_{jt} \sim \mathcal{N}(0,1)$ over the chunk's clean token activations x_t. Channel c clears if $\frac{1}{|R_i|}\sum_{r \in R_i} |\Delta^{(c)}_{i,r}| > q_{0.95}\{|\Delta^{(c)}_{u_j,r}| : j \le J, r \in R_i\}$.

Transfer. The statistic is $\max_f \mathrm{AUC}(a_f; S)$ over destination features; its null recomputes the same maximum on stratum-matched random k-sets, so the feature search is inside the null. A registered claim uses $\max(p, p_{\mathrm{rev}})$ over the two directions, Holm across claims.

Power. Before the private result is read, each claim's minimum detectable effect is simulated by shifting recentred per-series differences and re-testing with the same bootstrap (for example ΔMASE 0.133 at $n = 290$, 80% power, against an observed 0.209). The 20 transfer claims all reach the Holm floor. Single-feature causal claims are power-limited: each feature is tested on its 8 strongest-firing series, and 19 of the 21 v2 claims that did not confirm had not reached the dev p floor.

Benchmark and dictionaries. Every v2 sample passed the banded-DTW leakage gate (threshold 0.35; closest accepted sample 0.352), and 6 of 937,015 dev–private pairs are near-duplicates. Each target has three replicate dictionaries, and 9 of 11 atlas concepts are stable across them; held-out fraction of variance explained has medians of 0.647 (TimesFM) to 0.878 (Chronos-T5-Base), and fidelity does not predict causal clarity across v3 targets ($\rho = -0.326$, $n = 49$; MN-34). Retraining the v3 dictionaries from the recorded configuration reproduced all 50 exactly and the same 91 causal features.

Integrity of the confirmation. The v2 registry (128 hypotheses, sha256 recorded) was frozen before the private epoch was generated; the epoch shares no sample hash with any of seven earlier splits, and `confirm` ran once. Registered claims are cuts from larger dev pools (transfer 20 of 1,124 candidates, causal 32 of 113), which is legitimate because the private split referees them. The v3 registry (111 hypotheses) and the window-geometry claims were frozen in the same way before their epochs were extracted.

Appendix D. Reproducibility

`examples/panel7_v2` holds the configuration, the report and every stage's JSON and parquet artifacts plus the confirmation; `paper/figures/make_figures.py` regenerates all data figures. From `tsfm_lens/`, `python run.py --config configs/panel7_v2.yaml` reproduces the run. Environment: Python 3.12.13, numpy 2.1.0, PyTorch 2.12.0 (CUDA 12.9), transformers 4.57.6, timesfm 2.0.2, chronos-forecasting 2.3.1, zarr 2.18.7, scikit-learn 1.7.2. The v2 run took 21.1 hours on one RTX A5000.

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